Fiscal Year:	FY 2004	**Task Last Updated:**	FY 10/16/2004
PI Name:	Vander Wal, Randall L. Ph.D.		
Project Title:	Carbon Nanostructure: Its Evolution and its Impact upon Soot Growth and Oxidation		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Randall.L.VanderWal@nasa.gov, randy@rvander.grc.nasa.gov	**Fax:**	FY 216 433-3793
PI Organization Type:	NASA CENTER	**Phone:**	216 433-9065
Organization Name:	NASA Glenn Research Center		
PI Address 1:	2100 Brookpark Road		
PI Address 2:			
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135-3191	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	09/01/2000	**End Date:**	09/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Vander Wal , Randy	**Contact Phone:**	216-433-9065
Contact Email:	Randall.L.VanderWal@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
	The basic experimental approach will be to use one flame to synthesize soot with subsequent injection into a second flame. Variations of flame conditions and geometries will provide for a selection of soot processes by which the objectives can be realized. Optical methods will be used to measure soot concentration, pyrommetry for soot temperature and gas temperatures using thermocouples. Velocity information, critical for determining residence time, will be obtained by PIV while key radical concentrations, such as OH and O2 will be measured by diode laser absorption spectroscopy. Electron spectroscopic and microscopic techniques will be used to characterize the soot nanostructural evolution prior to and during growth, thermal and oxidative densification and oxidation. Analysis of the high-resolution transmission electron microscope (HRTEM) image data will quantify changes in the soot graphitization level and be used to correlate		

Task Description:	both the development of nanostructure upon growth, temperature and oxidation conditions. Combined with primary particle size measurements, the impact of nanostructure upon these processes will be assessed. To analyze soot nanostructure, assess its evolution, to synthesize and characterize fullerenes and fullerenic structures and investigate soot oxidation as a function of oxidant, temperature and nanostructure. Specific objectives will be to show that a) soot growth conditions affect the resulting nanostructure (graphitization level and fullerenic structures), b) the initial soot nanostructure will impact the rate of soot oxidation, and c) oxidation in turn will alter the soot nanostructure, the degree depending upon the oxidation conditions. A final objective will to demonstrate that soot nanostructure, both nascent and as altered by oxidation will affect subsequent soot (re)growth. Soot impacts the performance of power and propulsion systems, is a hazard of unwanted fires and is the main combustion generated pollutant. Responsible for an estimated 60,000 deaths per year in the United States alone, the Environmental Protection Agency (EPA)'s new Particulate Matter (PM) 2.5 criteria for particulate matter reflects increasing recognition of the health hazards associated with soot. Combustor ignitors and fuel injectors can become fouled with flame generated soot. Continuum radiation from soot is the main heat load upon combustor components and linings and is ultimately responsible for their failure. The spread of unwanted fires largely occurs through preheating from soot radiation. While an unwanted pollutant, soot also causes the release of substantial CO (carbon monoxide) emissions from fires and unburned hydrocarbons in engines. In contrast to seeking control of soot inception and growth, an alternative strategy is to focus on soot removal through oxidation. Current models for soot oxidation are inhibited by widely varying values for soot oxidation rates, differing by a factor of 20.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	By applying our fringe analysis algorithms to the soot samples from our collaborators, we believe we can use our results to both understand the combustion process by which the soot was formed, and identify the source of the soot.
Task Progress:	The progress below covers the duration of the grant. Title: Soot Oxidation: Dependence Upon Initial Nanostructure Conclusions The work reported here illustrates a dependence of the soot particle nanostructure upon synthesis conditions; namely temperature, time and initial fuel identity. Such structural variations in the graphene layer plane dimensions necessarily alters the ratio of basal plane versus edge site carbon atoms. A corresponding variation in the overall reactivity, reflecting an average of the different reactivities associated with these specific atomic sites arises. This variation is illustrated here between a disordered soot derived from benzene and a graphitic soot derived from acetylene. Their oxidation rates differ by nearly 5-fold. Curvature of layer planes, as observed for an ethanol derived soot, is found to substantially increase oxidative reactivity. Relative to fringe length as a manifestation of graphitic structure, curvature more effectively increases reactivity towards oxidation. Larger variations in oxidation behavior may be expected, depending upon the soot synthesis conditions. Other physical properties may similarly be affected. Title: Soot Nanostructure: Dependence Upon Synthesis Conditions Conclusions As the results show, the nanostructure of soot does indeed depend upon its formation conditions. Fuel identity governs the soot nanostructure under specific combinations of temperature and residence times as dictated by flow rates. At 1250 ºC, fuel structure effects are hidden, as all fuels, including polycyclic aromatic hydrocarbons (PAHs) tested produced an amorphous soot regardless of the flow rate. These results are interpreted as soot mass growth being dominated by PAHs. An increased flow rate results in soot nanostructure that is controlled largely by the pyrolysis kinetics for the pure hydrocarbon fuels. Variation of the flow rate alters the pyrolysis chemistry; specifically the species that add mass to the soot particle. These in turn are dependent upon the temperature. Title: Carbon Nanostructure Examined by Lattice Fringe Analysis of High Resolution Transmission Electron Microscopy Images In conclusion, a lattice fringe analysis program has been developed to quantify the data conveyed by HRTEM images. By providing a direct measure of the molecular level of graphitic structure, it is expected that its application could provide predictive capabilities for carbon reactivity, particularly towards oxidation. The robustness of this program is demonstrated by using a series of carbon blacks possessing different levels of graphitic structure, prepared at different heat treatment temperatures. Its credibility is benchmarked against a traditional measure of graphitic structure, as provided by Raman analysis. Lattice fringe length is found to be monotonic with the level of graphitic structure as provided by the ratio of the integrated intensities of the G/D spectral peaks in the Raman spectra. Reciprocally, for production purposes, the G/D band intensity ratio can serve as a direct measure of the lattice fringe reactivity. Title: Analysis of HRTEM Images for Carbon Nanostructure Quantification Conclusions Our image processing algorithm has been expanded to extract fringe separation and tortuosity in addition to fringe length from HRTEM images. An image analysis algorithm has been developed to extract the different measures of lattice fringe length, separation distance and tortuosity from HRTEM images. Validation of the lattice fringe separation distance has been tested against a series of heat-treated carbon blacks representing samples possessing different levels of graphitic structure. The results are compared to X-ray Powder Diffraction (XRD) data and found to be in reasonable agreement. The ability of this algorithm to further distinguish soots based on fringe length and tortuosity is tested using a series of soots synthesized by pyrolysis of hydrocarbon fuels at different flow rates and temperatures within a tube furnace. Amorphous soots produced at low temperatures are contrasted with those produced using high synthesis temperatures. Fringe length increases and interlayer spacing decreases with the observable increase in graphitic structure and order. Alteration of fuel or mass growth species (achieved by varying flow rate in the pyrolysis process) produces

	significant curvature within the carbon lamella. The curvature is shown to result in an increase in the fringe spacing. Tortuosity provides a measure of the curvature of the layer planes which further distinguishes the nanostructure of these carbons from those with graphitic structure. Acknowledgements This work was supported by a NASA NRA 99-HEDs-01 combustion award (RVW) administered through NASA cooperative agreement NAC3-975 with The National Center for Microgravity Research on Fluids and Combustion (NCMR) at The NASA-Glenn Research Center. The authors gratefully acknowledge Gordon M. Berger for assistance with the experiments and Dr. Y. L. Chen and David R. Hull for the TEM imaging. The authors gratefully acknowledge William K. Thompson (NASA-Glenn) for assistance with the software. Mr. James D. King and Michael I. Pamphlet acknowledge support through NASA-Glenn L.E.R.C.I.P. and Christina D. Taylor through the OAI accompanying student summer program.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Articles in Peer-reviewed Journals	Vander Wal RL, Tomasek AJ, Street K, Hull DR, Thompson WK. "Carbon nanostructure examined by lattice fringe analysis of high-resolution transmission electron microscopy images." Appl Spectrosc. 2004 Feb;58(2):230-7. https://doi.org/10.1366/000370204322842986 ; PubMed PMID: 17140483 , Feb-2004
Articles in Peer-reviewed Journals	Vander Wal RL, Tomasek AJ. "Soot nanostructure: dependence upon synthesis conditions." Combustion and Flame. 2004 January;136(Issues 1–2):129-40. https://doi.org/10.1016/j.combustflame.2003.09.008 , Jan-2004